



Girraween High School

Year 12 Mathematics Task 1

November 2016

General Instructions

- Working Time - 1 hour & 30 minutes
- Calculators and ruler may be used
- All necessary working out must be shown

Total Marks - 70

- Attempt all questions
- Marks may be deducted for careless or badly arranged work

Question 1 (1 mark)

Two six sided dice are thrown. What is the probability that the numbers appearing are the same?

- A. $\frac{1}{6}$ B. $\frac{1}{36}$ C. $\frac{1}{2}$ D. $\frac{1}{4}$

Question 2 (1 mark)

A geometric sequence with common ratio r will have a limiting sum if:

- A. $r < 1$ B. $r > 1$ C. $|r| < 1$ D. $|r| > 1$

Question 3 (1 mark)

The fourth term of an arithmetic series is 27 and the seventh term is 12. What is the common difference?

- A. 5 B. -5 C. 13 D. -13

Question 4 (1 mark)

The parabola with equation $(x - 3)^2 = 12y$ has:

- A. vertex at $(0, -3)$ and focus at $(3, -3)$
B. vertex at $(-3, 0)$ and focus at $(-3, 3)$
C. vertex at $(3, 0)$ and focus at $(3, 3)$
D. vertex at $(0, 3)$ and focus at $(3, 3)$

Question 5 (1 mark)

A bag contains 4 red marbles and 6 blue marbles. Three marbles are selected at random without replacement. What is the probability that at least one of the marbles selected is red?

- A. $\frac{1}{6}$ B. $\frac{1}{2}$ C. $\frac{5}{6}$ D. $\frac{29}{30}$

Question 6 (11 marks)

- (a) For the arithmetic sequence: 58, 53, 48, ...
- i. Find the values of a and d . [2]
 - ii. Find the expression for the n^{th} term. [2]
 - iii. Find the 30^{th} term. [1]
 - iv. Find the first negative term in the sequence. [2]
- (b) Consider the sequence 6, 11, 16, ..., 211.
- i. Determine the number of terms in this sequence. [2]
 - ii. Find the value of $6 + 11 + 16 + \dots + 211$ [2]

Question 7 (10 marks)

- (a) The 3^{rd} term of a geometric sequence is 8 and the 7^{th} term is 128.
- i. Find the 1^{st} term and the common ratio. [2]
 - ii. Find the sum of the first 20 terms. [1]
- (b) Using a limiting sum, express $0.\dot{2}\dot{5}$ as a fraction. [2]
- (c) For what values of x does the series $1 - 2x + 4x^2 - 8x^3 + \dots$ have a limiting sum? [2]
- (d) A ball is dropped from a height of 8m . At subsequent rebounds, it rises to a height of 80% of the preceding attained height. Find the total distance travelled (up and down) by the ball. [3]

Question 8 (12 marks)

For the parabola: $(x + 2)^2 = 8(y + 4)$

- i. Find the focal length [1]
- ii. Find the coordinate of the vertex [1]
- iii. Find the equation of the axis of symmetry [1]
- iv. Find the equation of the directrix [1]
- v. Find the x and y intercepts of the parabola [2]
- vi. Find the equation of the focal chord which passes through the point $(2, 5)$ [2]
- vii. Sketch the parabola showing the vertex, directrix, axis of symmetry, focus and the intercepts [4]

The exam continues on the next page

Question 9 (8 marks)

- (a) Given that a parabola has its focus at $(1, -1)$ and its directrix is given by $y = 3$, find:
- i. the focal length [1]
 - ii. the equation of the parabola [2]
- (b) A parabola has equation $x = \frac{y^2}{12} + \frac{y}{6} - \frac{23}{12}$. Find:
- i. the coordinate of its vertex [3]
 - ii. the coordinate of its focus [1]
 - iii. the equation of the directrix [1]

Question 10 (7 marks)

- (a) Find the locus of a point $P(x, y)$ such that P is equidistant from the x -axis and the point $F(0, -3)$. [3]
- (b) The locus of a point $P(x, y)$ which moves such that PA is always perpendicular to PB where $A(3, 4)$ and $B(-1, 2)$ is a circle. Find the centre and radius of this circle. [4]

Question 11 (9 marks)

- (a) Two six sided dice are thrown, find that probability that:
- i. the sum of the numbers on the two dice is 6 [1]
 - ii. the sum of the numbers on the two dice is less than 6 [1]
- (b) In a raffle, 20 tickets are sold and there are 2 prizes, first and second prize.
- i. What is the probability that someone buying 5 tickets wins the first prize? [1]
 - ii. What is the probability that someone buying 5 tickets wins at least one prize? [2]
- (c) Out of 27 students, 18 play volleyball and 12 play basketball.
- i. By drawing a venn diagram, state how many students play both sports. [3]
 - ii. If a student is chosen at random, what is the probability that the student plays either sport, but not both? [1]

The exam continues on the next page

Question 12 (8 marks)

- (a) A card is drawn from a standard deck of cards, and a six sided die is thrown.
- i. By drawing a probability tree, find the probability that the card is a heart and the number on the die is greater than 3. [2]
 - ii. Find the probability that the number on the card matches with the number on the die. (assuming ace corresponds to 1). [2]
- (b) Pam and Quin are playing a turn based game. The probability of Pam winning on any given turn is p and probability of Quin winning on any given turn is q . The probability of any given turn ending in a draw is r . The game continues until a winner is determined and Pam takes the first turn.
- i. Find the probability that Pam wins on her first or second turn. [1]
 - ii. Show that Pam's probability of winning this game is given by $\frac{p}{(p+q)(2-p-q)}$. [3]

End of exam

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MC: A, C, B, C, C.

Q1

$$p = 6 \times \frac{1}{6} \times \frac{1}{6} = \frac{1}{6} \therefore \textcircled{A}$$

Q2

$$|r| < 1 \therefore \textcircled{C}$$

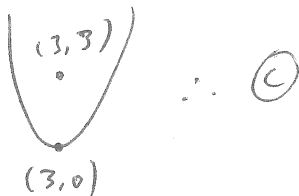
Q3

$$a + 3d = 27$$

$$a + 6d = 12$$

$$\therefore 3d = -15 \therefore d = -5 \therefore \textcircled{B}$$

Q4



Q5

$$= 1 - P(\text{all blue})$$

$$= 1 - \left(\frac{6}{10} \times \frac{5}{9} \times \frac{4}{8} \right) = \frac{5}{6} \therefore \textcircled{C}$$

Q6

(a)

$$(i) a = 58 \quad d = -5$$

$$(ii) T_n = 58 - 5(n-1)$$

$$= 58 - 5n + 5$$

$$T_n = 63 - 5n$$

(iii)

$$T_{30} = 63 - 5 \times 30$$

$$= -87$$

$$(iv) 63 - 5n < 0$$

$$5n > 63$$

$$n > \frac{63}{5}$$

$$n > 12.6$$

$$\therefore n = 13$$

$$T_{13} = 63 - 5 \times 13$$

$$= -2$$

(b)

$$(i) T_n = a + (n-1)d$$

$$211 = 6 + 5(n-1)$$

$$211 = 6 + 5n - 5$$

$$\therefore 5n = 210$$

$$\therefore n = 42$$

$\therefore 42$ terms in sequence.

$$(ii) S_{42} = \frac{42}{2} (6 + 211)$$

$$= 4557$$

Q7

(a) $8 = ar^2$

$128 = ar^6$

(i) $\therefore r^4 = \frac{128}{8} = 16$

$\therefore r = \pm 2$

$\therefore a = 2$

(ii) $S_{20} = \frac{2(2^{20}-1)}{2-1} \approx S_{20} = \frac{2(1-2^{20})}{1+2}$
 $= 2097150 \qquad = -699050$

(b) $0.\dot{2}\dot{5} = 0.252525\dots$

$= 0.25 + 0.0025 + 0.000025 + \dots$

$S_{\infty} = \frac{0.25}{1-0.01} = \frac{25}{99}$

(c)

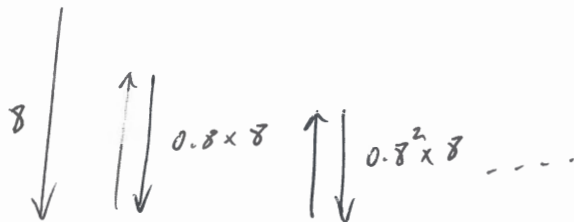
$r = -2n$

$\therefore -1 < -2n < 1$

$\therefore \frac{1}{2} > n > -\frac{1}{2}$

$\therefore -\frac{1}{2} < n < \frac{1}{2}$

(d)



$D = 8 + 2 \times 0.8 \times 8 + 2 \times 0.8^2 \times 8$
 $+ 2 \times 0.8^3 \times 8 + \dots$

$= 8 + 16(0.8 + 0.8^2 + 0.8^3 + \dots)$

$= 8 + 16\left(\frac{0.8}{1-0.8}\right)$

$= 72m$

Q8

$$(x+2)^2 = 8(y+4).$$

i. $4a = 8 \therefore a = 2$

ii. $V = (-2, -4)$

iii. $x = -2$

iv. $y = -6$

v. $y = 0$ when

$$(x+2)^2 = 32.$$

$$\therefore x+2 = \pm 4\sqrt{2}.$$

$$\therefore x = -2 \pm 4\sqrt{2}$$

$\therefore$ x -intercepts are $(-2+4\sqrt{2}, 0)$ &
 $(-2-4\sqrt{2}, 0)$

$x = 0$ when

$$4 = 8(y+4)$$

$$\therefore y+4 = \frac{1}{2} \therefore y = -3\frac{1}{2}.$$

$\therefore$ y -intercept is $(0, -3\frac{1}{2})$.

vi. $F = (-2, -2)$

$$m = \frac{5+2}{2+2} = \frac{7}{4}.$$

$$y+2 = \frac{7}{4}(x+2)$$

$$4y+8 = 7x+14$$

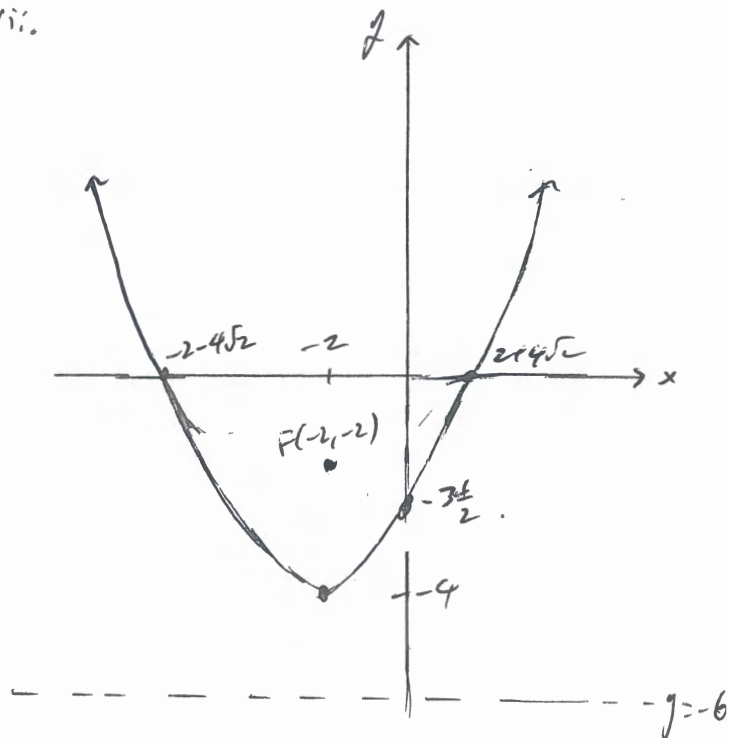
$$7x-4y+6=0.$$

or

$$4y = 7x+6$$

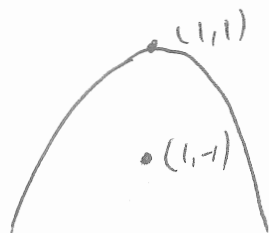
$$y = \frac{7}{4}x + \frac{3}{2}.$$

vii.



29

(a) $y = 3$



(i) $2a = 4 \therefore a = 2$

(ii) $(x-1)^2 = -8(y-1)$

(b)

(i) $x = \frac{y^2}{12} + \frac{y}{6} - \frac{23}{12}$

$12x = y^2 + 2y - 23$

$12x + 23 = y^2 + 2y$

$12x + 23 + 1 = y^2 + 2y + 1$

$12x + 24 = (y+1)^2$

$12(x+2) = (y+1)^2$

$\therefore V = (-2, -1)$

(ii) $a = 3$

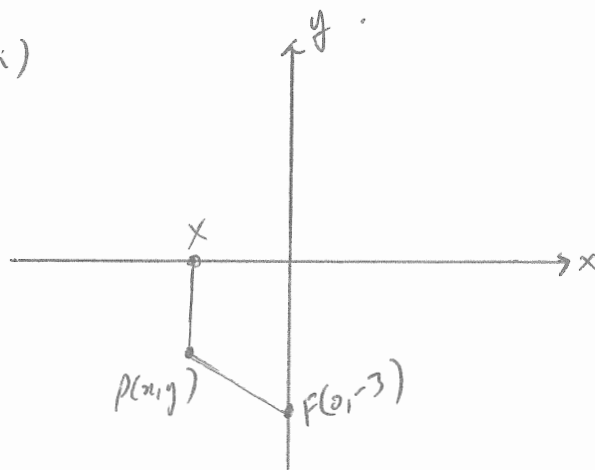


$\therefore F = (1, -1)$

(iii) $x = -5$

210

(a)



$PX = PF$

$\therefore -y = \sqrt{x^2 + (y+3)^2}$

$y^2 = x^2 + (y+3)^2$

$x^2 = y^2 - (y+3)^2$

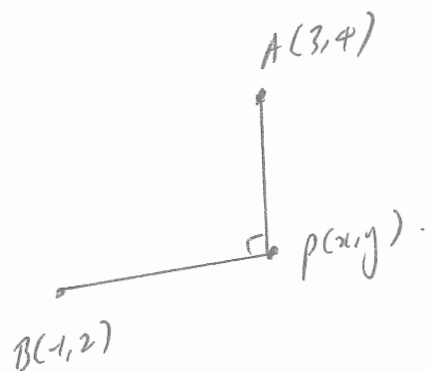
$x^2 = (y-y-3)(y-y+3)$

$x^2 = -3(2y+3)$

$x^2 = -6(y + \frac{3}{2})$

Q13

(b)



$$m_{PA} = \frac{y-4}{x-3} \quad m_{PB} = \frac{y-2}{x+1}$$

$$m_{PA} \times m_{PB} = -1$$

$$\therefore \frac{y-4}{x-3} \times \frac{y-2}{x+1} = -1$$

$$\therefore (y-4)(y-2) = (3-x)(x+1)$$

$$y^2 - 2y - 4y + 8 = 3x + 3 - x^2 - x$$

$$x^2 - 2x + y^2 - 6y = -5$$

$$x^2 - 2x + 1 + y^2 - 6y + 9 = -5 + 1 + 9$$

$$(x-1)^2 + (y-3)^2 = 5$$

$$\therefore C = (1, 3) \quad r = \sqrt{5}$$

Q11

(a)

(i)	1	2	3	4	5	6
1					X	
2				X		
3			X			
4		X				
5	X					
6						

$$\therefore p = \frac{5}{36}$$

$$(ii) \quad p = \frac{10}{36} = \frac{5}{18}$$

(b)

$$(i) \quad \frac{5}{20} = \frac{1}{4}$$

$$(ii) \quad p = 1 - p(\text{no prize})$$

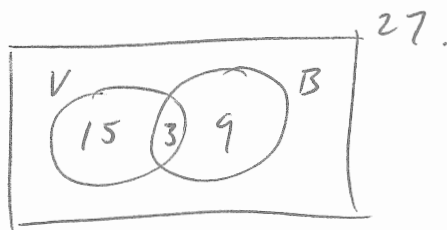
$$= 1 - \frac{15}{20} \times \frac{14}{19}$$

$$= \frac{17}{38}$$

(c)

$$(i) \quad 18 - x + x + 12 - x = 27$$

$$\therefore x = 3 \quad \therefore 3 \text{ plays both.}$$



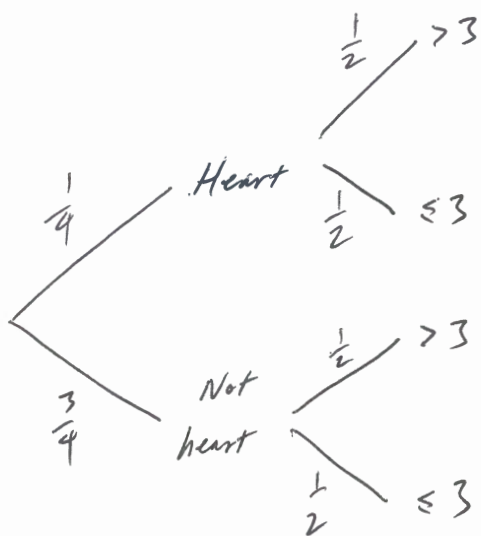
(ii)

$$p = \frac{15+9}{27} = \frac{24}{27} = \frac{8}{9}$$

Q12

(a)

(i)



$$p(\text{heart \& } >3) = \frac{1}{8}$$

$$(ii) \quad p(\text{ace \& } 1) = \frac{4}{52} \times \frac{1}{6}$$

$$\therefore p(\text{same number}) = 6 \times \frac{1}{6} \times \frac{4}{52} \\ = \frac{1}{13}$$

(b)

(i)

$$p = p + r^2 p$$

(ii)

$$p_{\text{win}} = p + r^2 p + r^4 p + r^6 p + \dots \\ = p(1 + r^2 + r^4 + r^6 + \dots) \\ = p \times \frac{1}{1-r^2} \\ = \frac{p}{1-r^2}$$

$$\text{But } p+q+r=1$$

$$\therefore r = 1-p-q$$

$$\therefore 1-r^2 = (1-r)(1+r) \\ = (p+q)(2-p-q)$$

$$\therefore p_{\text{win}} = \frac{p}{(p+q)(2-p-q)}$$